

Exp 2

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Code :

Candidate Elimination Algorithm without CSV

```
data = [
    ['High', 'High', 'High', 'Yes'],
    ['High', 'High', 'Medium', 'Yes'],
    ['High', 'Medium', 'Medium', 'Yes'],
    ['Medium', 'Medium', 'Medium', 'No'],
    ['Low', 'Low', 'Low', 'No'],
    ['High', 'High', 'High', 'Yes'],
    ['Medium', 'Medium', 'Medium', 'No'],
    ['High', 'High', 'Medium', 'Yes'],
    ['Low', 'Low', 'Low', 'No'],
    ['High', 'High', 'High', 'Yes']
]

# Initialize Specific Hypothesis
S = data[0][: -1]

# Initialize General Hypothesis
G = [['?' for i in range(len(S))]]

print("Initial Specific Hypothesis:")
print(S)

print("\nInitial General Hypothesis:")
print(G)

for sample in data:
    if sample[-1] == "Yes":
        for i in range(len(S)):
            if sample[i] != S[i]:
                S[i] = '?'
                G[0][i] = '?'
    else:
        for i in range(len(S)):
            if sample[i] != S[i]:
                temp = ['?' for j in range(len(S))]
                temp[i] = S[i]
```

```

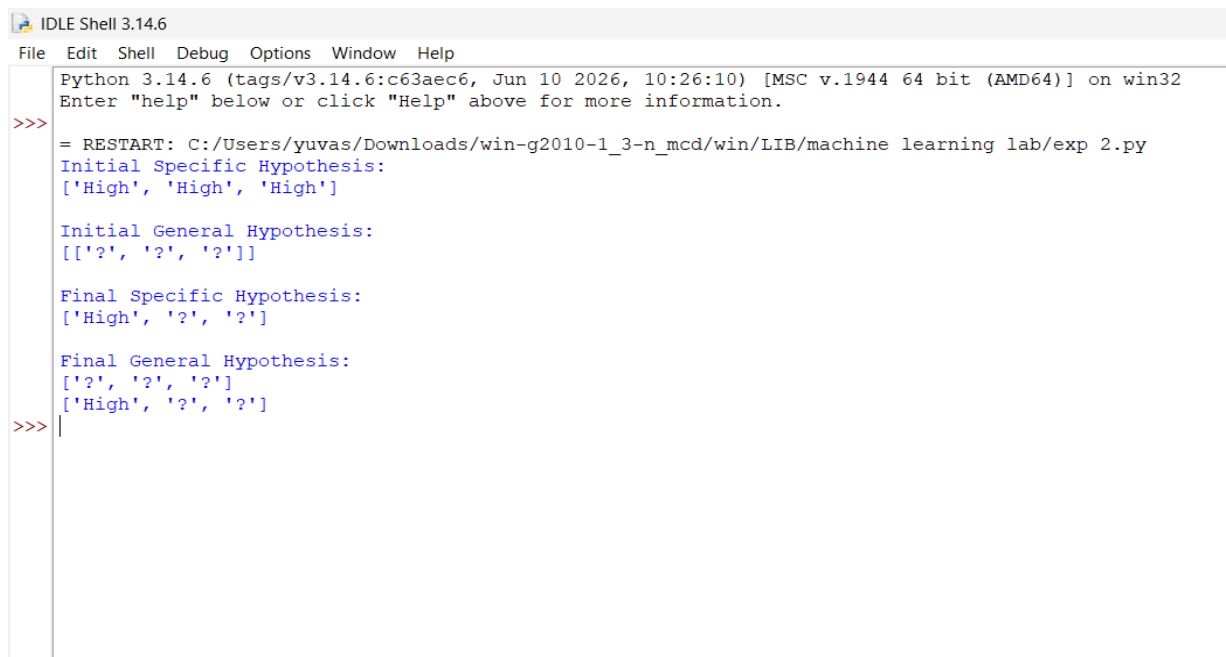
        if temp not in G:
            G.append(temp)

print("\nFinal Specific Hypothesis:")
print(S)

print("\nFinal General Hypothesis:")
for g in G:
    print(g)

```

Output:



```

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/exp 2.py
Initial Specific Hypothesis:
['High', 'High', 'High']

Initial General Hypothesis:
[['?', '?', '?']]

Final Specific Hypothesis:
['High', '?', '?']

Final General Hypothesis:
[['?', '?', '?']]
['High', '?', '?']
>>> |

```